

COMPUTER NETWORKS LAB**Course Code: IT621****Credit Unit: 01****Total Hours: 30****Course Objective:**

To design and execute models in packet tracer to demonstrate different applications of computer networks such as addressing, routing, layer2 switching, layer3 switching, VTP, VLAN, Routing tables and Routing Algorithms, Frame Relay GRE, MPLS, EtherChannel and Tunneling. To design and execute codes in a simulated environment to implement various Routing and Switching methods.

SOFTWARE REQUIREMENTS: Packet Tracer or GNS**Course Contents:**

Lab Experiments are based on the course Data Communication and Computer Networks (IT601)

List of experiments/ demonstrations: (Each experiment is of 2 Hours duration)

- 1 Study of different types of cables and connectors and Network devices, display topologies demonstrating different types of devices and connectors (2 Hrs.)
- 2 Demonstrate the differences between connecting LAN and WAN devices and design LAN with WAN Topologies (2 Hrs.)
- 3 Study of basic network commands and network configuration commands (2 Hrs.)
- 4 Configure different types of VLAN using the simulator (2 Hrs.)
- 5 Configure VTP for different types of VLAN using the simulator (2 Hrs.)
- 6 Configure STP for different types of VLAN using the simulator (2 Hrs.)
- 7 Configure Frame Relay for different types of WAN using the simulator (2 Hrs.)
- 8 Configure IPSec for different types of WAN using the simulator (2 Hrs.)
- 9 Configure RIP, RIPv2 protocols using the simulator (2 Hrs.)
- 10 Configure EIGRP for IPv4 and IPv6 protocols using the simulator (2 Hrs.)
- 11 Configure OSPFv2 and OSPFv3 protocols using the simulator (2 Hrs.)
- 12 Differentiate Inter and Intra-Domain and WAN Routing protocols using the simulator (2 Hrs.)
- 13 Configure MPLS for WAN using the simulator (2 Hrs.)
- 14 Configure GRE Tunneling using the simulator (2 Hrs.)
- 15 Configure PAP, CHAP, EtherChannel: Static, LACP, PAGP using the simulator (2 Hrs.)

Course Outcomes:

- Ability to identify different network cables and logical topology problems
- Ability to provide network design solutions.
- Ability to design and analyze different routing and switching methods.
- Ability to effectively use WAN and LAN design methods using simulators.

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva



K. P. Jaiswal
Director-ASET
Amity University Madhya Pradesh Gwalior

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Note: IA – Internal Assessment, EE- External Exam, A- Attendance, PR- Performance, LR – Lab Record, V – Viva.

Text & References:

- CCNA. A Comprehensive Beginners Guide To Learn About The CCNA Routing And Switching, Pearson VUE
- Cisco CCNA Command Guide, TMH
- Comptia Network+ Study Guide, Todd Lammie, TMH